

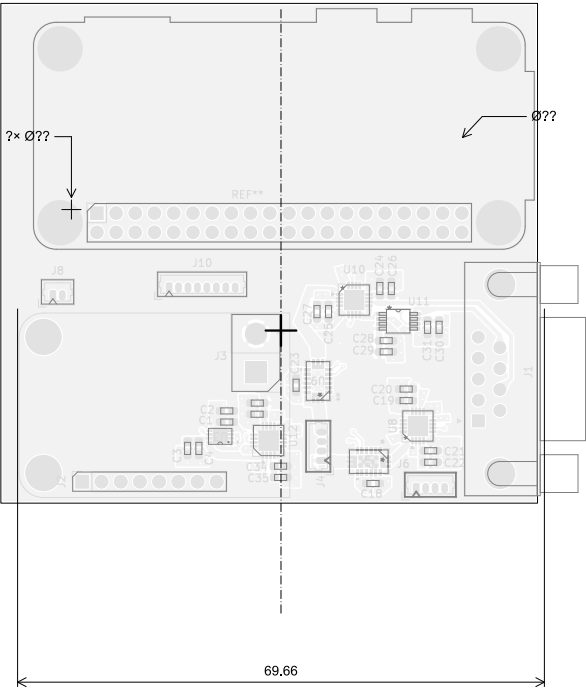
alb_mlb test Fabrication Document

Layer Stack Legend

Material	Layer	Thickness	Dielectric	Type	Gerber
F.Paste				Paste Mask	
F.Silkscreen			Direct Printing	Legend	GBR
F.Mask		0.02mm	Solder Resist	Solder Mask	GBR
Copper	L1 (Sig, PWR)	0.07mm (2.90oz)		Signal	GBR
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	L2 (GND)	0.035mm (1.00oz)		Plane	GBR
Core		1.21mm	FR4	Dielectric	
Copper	L3 (Sig, PWR)	0.035mm (1.00oz)		Signal	GBR
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	L4 (Sig, PWR)	0.07mm (2.90oz)		Signal	GBR
B.Mask		0.02mm	Solder Resist	Solder Mask	GBR
B.Silkscreen			Direct Printing	Legend	GBR
B.Paste				Paste Mask	

Total thickness: 1.66mm
Note: external layer thicknesses are specified after plating

Top Fabrication (Scale 1:1)



Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0.2032	0.28	L2

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING YELLOW NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

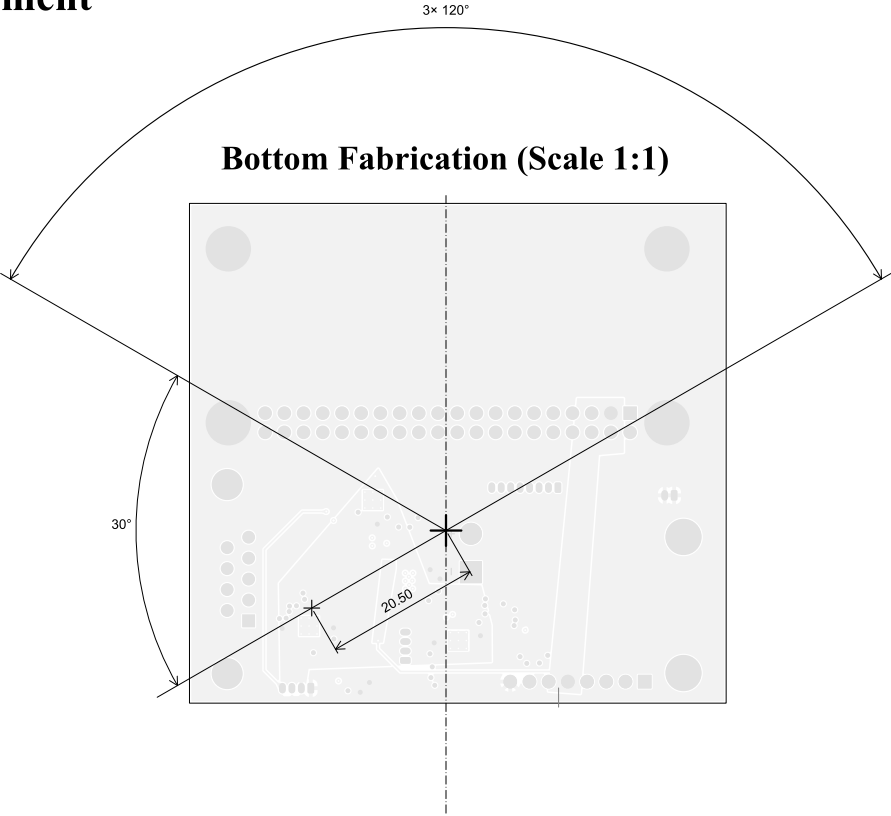
A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
B. Tg 170 C OR EQUIVALENT.
C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY GENERAL ATOMICS EMS.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE70.968 × 66.091 mm
BOARD THICKNESS1.660 mm
TRACE WIDTH0.200 mm
TRACE TO TRACE0.200 mm
MIN. HOLE (PTH)0.200 mm
MIN. HOLE (NPTH)2.750 mm
ANNULAR RING0.150 mm
COPPER TO HOLE0.254 mm
COPPER TO EDGE0.250 mm
HOLE TO HOLE0.254 mm
- 11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.
- 12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

	Comments:	Company: General Atomics EMS		Variant: PRELIMINARY	Git Hash: 791ce63
	Sheet Title: Top Fabrication (Scale 1:1)	Board Name: alb_mlb test		Project Name: ALB/MLB test	
	Sheet Path:	File Name: proalbmblb.kicad_pcb	Designer: Marcus Reighter	Date: 2024-04-13	Revision: + (Unreleased)
			Reviewer:	Size: A4	Sheet: 1 of 14

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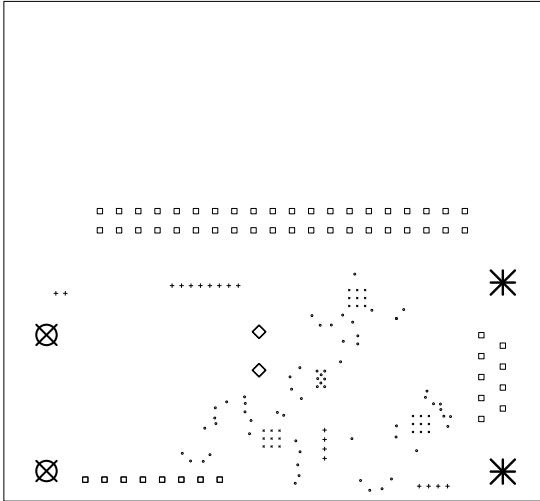
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	Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: proalbmlb.kicad_pcb	Designer: Marcus Reighter	Date: 2024-04-13	Revision: + (Unreleased)
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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	27	0.20mm (7,87mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
○	63	0.25mm (9,84mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Via
+	18	0.50mm (19,69mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
□	65	1.00mm (39,37mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
◇	4	1.75mm (68,90mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
⊗	2	2.70mm (106,30mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
⋈	2	3.20mm (125,98mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
Total 181						

Drill Drawing L1 - L4 (Scale 1:1)



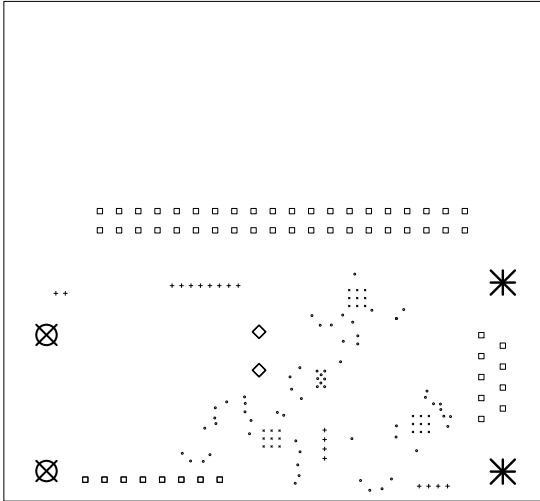
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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	27	0,20mm (7,87mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
○	63	0,25mm (9,84mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Via
+	18	0,50mm (19,69mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
□	65	1,00mm (39,37mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
◇	4	1,75mm (68,90mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
⊗	2	2,70mm (106,30mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
⌘	2	3,20mm (125,98mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
Total 181						

Drill Drawing L1 - L4 (Scale 1:1)



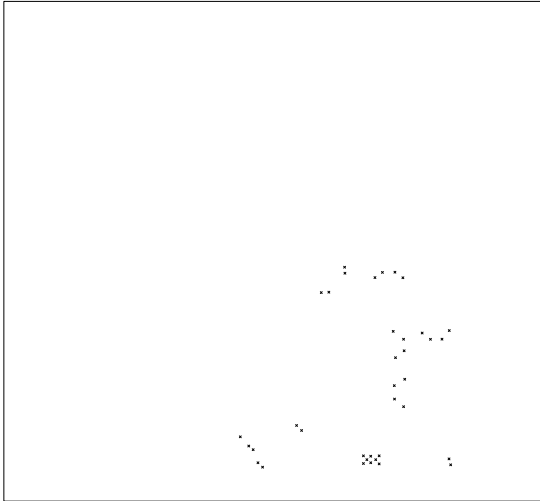
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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	37	0.25mm (9.84mils)	PTH	Round	L1 (Sig, PWR) - L2 (GND)	Via
	Total 37					

Drill Drawing L1 - L2 (Scale 1:1)



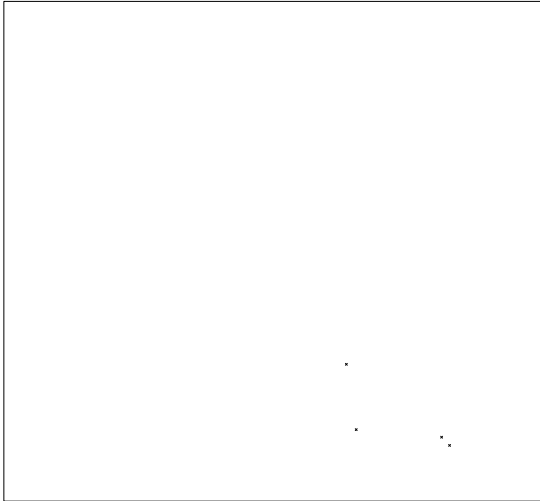
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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	4	0.25mm (9.84mils)	PTH	Round	L1 (Sig. PWR) - L3 (Sig. PWR)	Via
	Total 4					

Drill Drawing L1 - L3 (Scale 1:1)



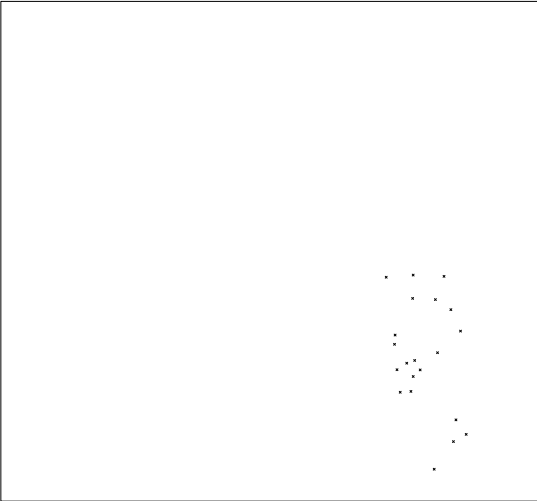
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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	21	0.25mm (9.84mils)	PTH	Round	L2 (GND) - L3 (Sig, PWR)	Via
	Total 21					

Drill Drawing L2 - L3 (Scale 1:1)



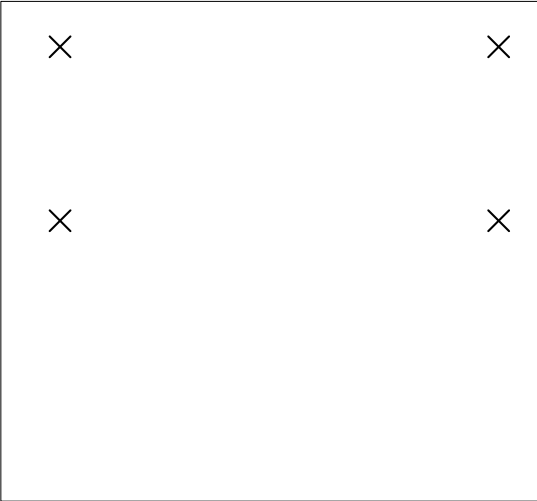
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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	4	2.75mm (108,27mils)	NPPTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Mechanical
	Total 4					

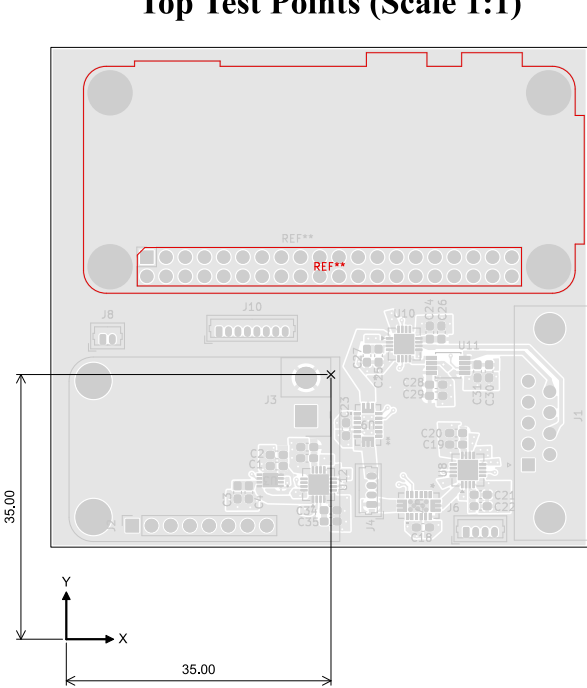
Drill Drawing L1 - L4 (Scale 1:1)



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	Sheet Title: Drill Drawing (L1 - L4)	File Name: proalbmblb.kicad_pcb	Designer: Marcus Reighter	Date: 2024-04-13	Revision: + (Unreleased)
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Top Test Points (Scale 1:1)



Ref.	Net	X [mm]	Y [mm]
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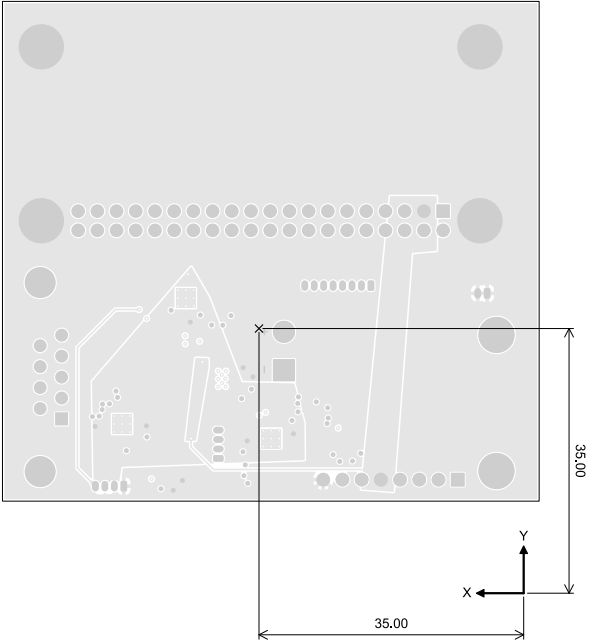
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Bottom Test Points (Scale 1:1)



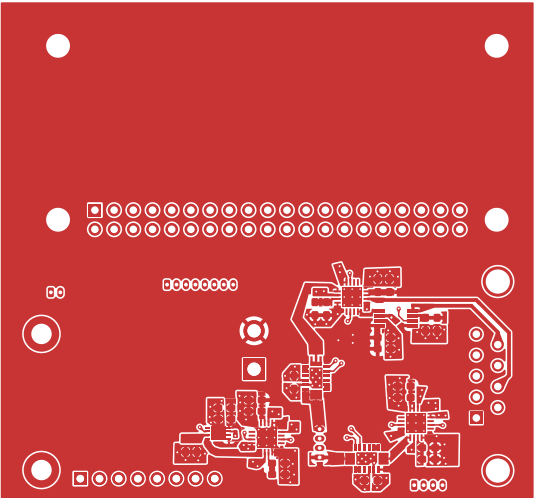
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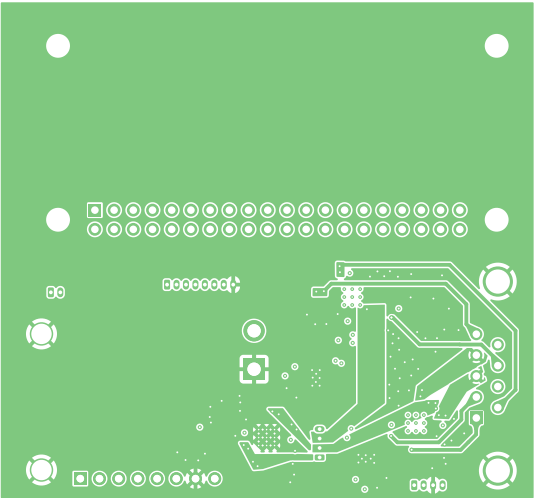
L1 (Sig, PWR) (Scale 1:1)



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	Sheet Title: L1 (Sig, PWR) (Scale 1:1)	File Name: proalbmlb.kicad_pcb	Designer: Marcus Reighter	Date: 2024-04-13	Revision: + (Unreleased)
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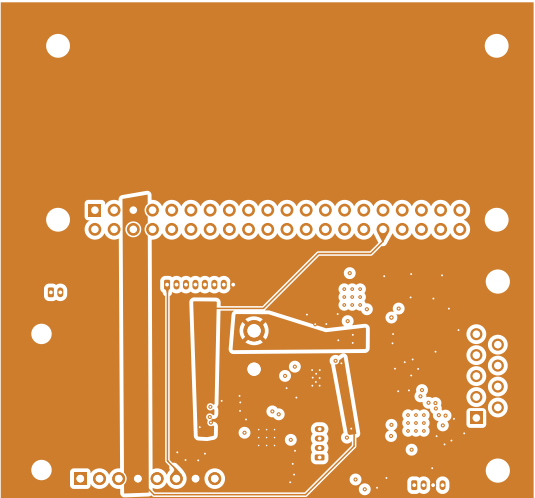
L2 (GND) (Scale 1:1)



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	Sheet Title: L2 (GND) (Scale 1:1)	File Name: proalbmlb.kicad_pcb	Designer: Marcus Reighter	Date: 2024-04-13	Revision: + (Unreleased)
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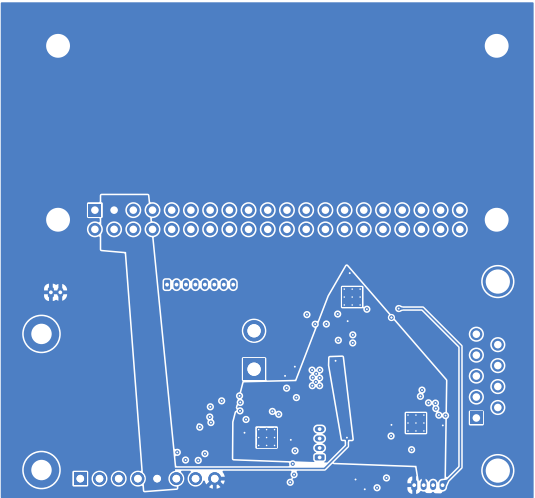
L3 (Sig, PWR) (Scale 1:1)



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L4 (Sig, PWR) (Scale 1:1)



	Comments:	Company: General Atomics EMS		Variant: PRELIMINARY	Git Hash: 791ee63
		Board Name: alb_mlb test		Project Name: ALB/MLB test	
	Sheet Title: L4 (Sig, PWR) (Scale 1:1)	File Name: proalbmlb.kicad_pcb	Designer: Marcus Reighter	Date: 2024-04-13	Revision: + (Unreleased)
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